Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 4 — The binomial expansion Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第4章 · 二项式展开
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P2-C04-R01 · P2 · Chapter 4

$f(x) = (1 + 2x)^6$.

(a) Find the first four terms in ascending powers of x. (3)

(b) Find the coefficient of x^3 in $(1 - x)f(x)$. (2)

(Total for Question 1 is 5 marks)

The coefficient of x^2 in $(1 + ax)^5$ is 90, where $a > 0$.

(a) Find a. (3)

(b) Find the coefficient of x^4 for this value of a. (2)

(Total for Question 2 is 5 marks)

The expansion of $(2+x)^n$ has constant term 32, where n is a positive integer.

(a) Find n. (2)

(b) For this value of n, find

(i) the coefficient of x^2, (2)

(ii) the coefficient of x^5 in $(2+x)^n(1+3x)$. (2)

(Total for Question 3 is 6 marks)

In the expansion of $(1+x)^n$, where n is an integer and $n \geq 3$, the coefficients of x, x^2 and x^3, in that order, form an arithmetic sequence.

(a) Show that $n^2 - 9n + 14 = 0$. (3)

(b) Hence

(i) find n. (2)

(ii) find the coefficient of x^4 in the expansion. (2)

(Total for Question 4 is 7 marks)

$D = (3 + x)^5 - (3 - x)^5$.

(a) Find D as a polynomial in x. (3)

(b) Hence find the coefficient of x^2 in D/x for $x \neq 0$. (2)

(Total for Question 5 is 5 marks)

6.

The first three terms of $(1 + px)^n$ are $1 + 12x + 60x^2$, where n is a positive integer and $p > 0$.

(a) Form two equations in n, p. (2)

(b) Find n, p. (3)

(Total for Question 6 is 5 marks)

7.

Consider $(1 + 0.02)^5$.

(a) Using the first three terms of a binomial expansion, estimate its value. (3)

(b) Find the exact sum of the omitted terms and state whether the estimate is too large or too small. (3)

(Total for Question 7 is 6 marks)

8.

The coefficient of x in $(3 + bx)^4$ is -216.

(a) Find b. (2)

(b) Find the coefficient of x^3 and explain its sign. (3)

(Total for Question 8 is 5 marks)

9. P2-C04-R09 · P2 · Chapter 4

$F(x) = (1 + x^2)^5$.

(a) Find the coefficient of x^6. (2)

(b) Find the coefficient of x^7 in $(2 + x)F(x)$. (3)

(Total for Question 9 is 5 marks)

The expansion of $(1+2x)^7$ is $a_0 + a_1x + \cdots + a_7x^7$.

(a) Find $a_0 + a_1 + \cdots + a_7$ without calculating the eight coefficients separately. (2)

(b) Find $a_0 - a_1 + a_2 - a_3 + \cdots - a_7$. (2)

(Total for Question 10 is 4 marks)

11.

In $(1 + x)^n$, the coefficient of x^2 is three times the coefficient of x.

(a) Find the positive integer n. (3)

(b) Find the coefficient of x^5 for this value of n. (2)

(Total for Question 11 is 5 marks)

$F(x) = (2 - x)^6$.

(a) Find the coefficients of x^2 and x^3. (2)

(b) Find k such that the coefficient of x^3 in $(1 + kx)F(x)$ is zero. (2)

(Total for Question 12 is 4 marks)

A cube has side $(10 + h)$ cm, where $h > 0$.

(a) Expand its volume in ascending powers of h. (2)

(b) For $h = 0.1$, find the error if only the constant and linear terms are used to estimate the volume. (3)

(Total for Question 13 is 5 marks)

The expansion of $(a + x)^5$, with $a > 0$, has coefficient of x^3 equal to 40.

(a) Find a. (2)

(b) Find the ratio of the coefficients of x^4 and x^2, in that order. (3)

(Total for Question 14 is 5 marks)

$P(x) = (1 + 3x)^4(1 - x)^2$.

(a) Find the constant and linear terms in $P(x)$. (2)

(b) Find the coefficient of x^2. (3)

(Total for Question 15 is 5 marks)

For a positive integer n, $(1-x)^n = 1 - 9x + ax^2 + bx^3 + \cdots$.

(a) Find n, a, b. (3)

(b) Find the coefficient of x^2 in $(1+x)(1-x)^n$. (2)

(Total for Question 16 is 5 marks)

Consider $(x + 2/x)^6$ for $x \neq 0$.

(a) Find the term independent of x. (3)

(b) Find the coefficient of x^2. (2)

(Total for Question 17 is 5 marks)

18.

The coefficient of x^2 in $(1+ax)^4+(1-ax)^4$ is 48, where $a>0$.

(a) Find a. (2)

(b) For this value, find the coefficient of x^4 and explain why the coefficient of x^3 is zero. (3)

(Total for Question 18 is 5 marks)

$F(x) = (3 - 2x)^5$.

(a) Find the coefficient of x^4. (2)

(b) Find the sum of all coefficients of the polynomial $F(2x)$. (3)

(Total for Question 19 is 5 marks)

The polynomial $P(x) = (1 + x)^6$ has coefficients a_k at x^k.

(a) Find a_2 and a_4 and explain why they agree. (3)

(b) Find the coefficient of x^3 in $(1 - x^2)P(x)$. (2)

(Total for Question 20 is 5 marks)

The expansion of $(2 + ax)^3$ has equal coefficients of x and x^2, where $a \neq 0$.

(a) Find a. (3)

(b) For this value of a, find the sum of the constant term and the coefficient of x^3. (2)

(Total for Question 21 is 5 marks)

Use the binomial theorem on $(100 - 1)^4$.

(a) Write the full expansion without evaluating 99^4 directly. (3)

(b) Hence find the final three digits of 99^4, explaining why the first three terms do not affect them. (3)

(Total for Question 22 is 6 marks)

$P(x) = (1 + x)^5(1 - x)^5$.

(a) Simplify $P(x)$ before expanding. (2)

(b) Find the coefficients of x^4 and x^7. (3)

(Total for Question 23 is 5 marks)

The first three terms of $(a + bx)^4$ are $16 + 96x + cx^2$, where $a > 0$.

(a) Find a, b. (3)

(b) Find c and the coefficient of x^4. (3)

(Total for Question 24 is 6 marks)

$F(x) = (1 - 2x)^7$.

(a) Find the first three non-zero terms in ascending powers of x. (3)

(b) Use these terms to estimate 0.98^7, and give the estimate to 4 decimal places. (3)

(Total for Question 25 is 6 marks)

In the expansion of $(x+3)^5$,

(a) find the coefficient of x^2, (2)

(b) find all real k such that the coefficient of x^2 in $(x+k)(x+3)^5$ is zero. (3)

(Total for Question 26 is 5 marks)

For a positive integer n, the coefficients of x and x^2 in $(1+x)^n$ differ by 14, with the coefficient of x^2 larger.

(a) Form and solve an equation for n. (4)

(b) Find the coefficient of x^3 for this value. (2)

(Total for Question 27 is 6 marks)

$P(x) = (1 + 2x)^4 + (1 - 2x)^4 - 2.$

(a) Find $P(x)$ in fully simplified form. (3)

(b) Show that $P(x) > 0$ for every real $x \neq 0$. (3)

(Total for Question 28 is 6 marks)

The expansion of $(1 + ax)^5$, where $a > 0$, has coefficient of x^2 equal to twice the coefficient of x.

(a) Find a. (2)

(b) For this value of a, the expansion of $(1 + ax)^5(1 + bx)$ has no x^3 term.

(i) Find b. (2)

(ii) Find the coefficient of x^5 in the product. (2)

(Total for Question 29 is 6 marks)

A polynomial $P(x)$ is defined by $P(x) = (2 + x)^6 + (2 - x)^6$.

(a) Explain why $P(x)$ contains no odd powers of x, and find its first three non-zero terms in ascending powers of x. (3)

(b) Let $Q(x) = (2 + x)^6 - (2 - x)^6$.

(i) Find the coefficient of x^3 in $Q(x)$. (2)

(ii) Find the coefficient of x^4 in $P(x) + xQ(x)$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C04-R01 · 5 marks

(a) · 3 marks

$1+12x+60x^2+160x^3$.

用二项式系数1、6、15、20，与(2x)^k相乘。

M1	正确二项式系数
M1	处理2x的各次幂
A1	四项

(b) · 2 marks

100.

所需项为原三次项减原二次项，160−60=100。

M1	识别两个贡献
A1	系数

Question 2

P2-C04-R02 · 5 marks

(a) · 3 marks

$a = 3$.

$10a^2=90$，正值取3。

M1	二次项系数关系
M1	正值选根
A1	a

(b) · 2 marks

405.

系数$5a^4=5\times 81$。

M1	正确项系数
A1	值

Question 3

P2-C04-R03 · 6 marks

(a) · 2 marks

$n = 5$.

常数项2^n=32，得5。

M1	常数项关系
A1	n

(b) For this value of n, find

(i) · 2 marks

80.

C(5,2)2^3=10×8。

M1	正确二项式项
A1	系数

(ii) · 2 marks

31.

来自原x^5系数1加3倍原x^4系数10，合31。

M1	两项贡献
A1	结果

Question 4

P2-C04-R04 · 7 marks

(a) · 3 marks

$n^2 - 9n + 14 = 0$.

三系数为$n, n(n-1)/2, n(n-1)(n-2)/6$。等差条件为$n(n-1) = n + n(n-1)(n-2)/6$；除以$n > 0$并整理得所求。

M1	三个二项系数
M1	等差条件
A1	代数整理

(b) Hence

(i) · 2 marks

$n = 7$.

$(n-2)(n-7) = 0$，由$n \geq 3$排除2。

M1	分解并用范围
A1	n

(ii) · 2 marks

35.

代入$n = 7$，系数$\binom{7}{4} = 35$。

M1	对应二项系数
A1	值

Question 5

P2-C04-R05 · 5 marks

(a) · 3 marks

$D = 810x + 180x^3 + 2x^5$.

偶次项抵消；奇次项两倍，$2\mathrm{C}(5,1)3^4=810$，$2\mathrm{C}(5,3)3^2=180$。

M1	奇偶项筛选
M1	正确系数与3的幂
A1	三项

(b) · 2 marks

180.

除x使幂次全降1，原三次系数成为二次系数。

M1	正确幂次对应
A1	系数

Question 6

P2-C04-R06 · 5 marks

(a) · 2 marks

$np=12,\ n(n-1)p^2/2=60.$

一次项与二次项分别比较。

M1	一次项条件
A1	二次项条件

(b) · 3 marks

$n=6, p=2.$

将第二式除第一式平方得(n−1)/(2n)=60/144=5/12，解n=6再求p。

M1	正确消去p
M1	解n
A1	两参数

Question 7

P2-C04-R07 · 6 marks

(a) · 3 marks

$1 + 0.10 + 0.004 = 1.104$.

取$1+5u+10u^2$，$u=0.02$。

M1	正确前三项
M1	代小量
A1	估计

(b) · 3 marks

0.0000808032; underestimate.

余项$10u^3+5u^4+u^5=0.00008+0.0000008+0.0000000032$，全正。

M1	列出全部余项
M1	计算余项
A1	数值及方向

Question 8

P2-C04-R08 · 5 marks

(a) · 2 marks

$b = -2$.

一次系数$4\times 3^{3}\mathrm{b}=108\mathrm{b}$。

M1	正确系数关系
A1	b

(b) · 3 marks

-96; it contains $b^{3} < 0$.

$\mathrm{C}(4,3)3\mathrm{b}^{3}=12\times(-8)$；其余因子正。

M1	正确三次项
A1	系数
B1	负号理由

Question 9

P2-C04-R09 · 5 marks

(a) · 2 marks

10.

需选三次x^2，$C(5,3)=10$。

M1	正确指数索引
A1	系数

(b) · 3 marks

10.

F只有偶次幂，2F没有x^7项，xF的x^7来自F中x^6。

M1	排除不可能幂次
M1	唯一贡献
A1	系数

Question 10

P2-C04-R10 · 4 marks

(a) · 2 marks

2187.

令$x=1$，系数和$=(1+2)^7=3^7$。

M1	选择$x=1$
A1	结果

(b) · 2 marks

-1.

令$x=-1$，值$=(1-2)^7=-1$。

M1	选择x负1
A1	结果

Question 11

P2-C04-R11 · 5 marks

(a) · 3 marks

$n = 7$.

n(n−1)/2=3n，n>0可除n，n−1=6。

M1	系数关系
M1	合法约去n
A1	n

(b) · 2 marks

21.

C(7,5)=C(7,2)=21。

M1	正确组合数
A1	系数

Question 12

P2-C04-R12 · 4 marks

(a) · 2 marks

240 and -160.

$C(6,2)2^4=240$；$C(6,3)2^3(-1)^3=-160$。

M1	正确二项式项
A1	两系数

(b) · 2 marks

$k=2/3$.

三次系数为$-160+240k$，令零。

M1	两个贡献并令零
A1	k

Question 13

P2-C04-R13 · 5 marks

(a) · 2 marks

$1000+300h+30h^2+h^3$ cm^3.

用三次二项式展开，系数1、3、3、1。

M1	正确二项式展开
A1	四项及单位

(b) · 3 marks

0.301 cm^3, underestimate.

遗漏项$30h^2+h^3=0.3+0.001$，均正。

M1	正确遗漏项
M1	代值
A1	误差及方向

Question 14

P2-C04-R14 · 5 marks

(a) · 2 marks

$a = 2$.

$\binom{5}{3}a^2=10a^2=40$，正值取2。

M1	系数条件
A1	a

(b) · 3 marks

$1:8$.

系数分别$5a=10$及$10a^3=80$，约简10:80。

M1	两个正确系数
M1	按顺序取比
A1	比值

Question 15

P2-C04-R15 · 5 marks

(a) · 2 marks

$1+10x$.

常数1；一次项12x−2x=10x。

M1	正确低次展开
A1	两项

(b) · 3 marks

31.

贡献54−24+1=31，分别来自二次×常数、一次×一次、常数×二次。

M1	列齐三种贡献
M1	正确三数
A1	系数

Question 16

P2-C04-R16 · 5 marks

(a) · 3 marks

$n=9, a=36, b=-84.$

一次项−n=−9，二次C(9,2)=36，三次−C(9,3)=−84。

M1	一次项确定n
M1	后续系数
A1	三值

(b) · 2 marks

27.

原二次36加x乘原一次−9，合27。

M1	两项贡献
A1	系数

Question 17

P2-C04-R17 · 5 marks

(a) · 3 marks

160.

第k次选2/x后幂为6−2k；零幂k=3，系数$C(6,3)2^3$。

M1	正确幂次表达式
M1	求索引
A1	常数项

(b) · 2 marks

60.

6−2k=2给k=2，系数$C(6,2)2^2$。

M1	幂次索引
A1	系数

Question 18

P2-C04-R18 · 5 marks

(a) · 2 marks

$a = 2$.

两展开式二次系数各$6a^2$，故$12a^2=48$。

M1	偶次项合并
A1	正a

(b) · 3 marks

32; the cubic terms cancel because their signs are opposite.

四次为$2a^4=32$；三次系数$4a^3$与$-4a^3$抵消。

M1	四次贡献
A1	32
B1	三次抵消理由

Question 19

P2-C04-R19 · 5 marks

(a) · 2 marks

240.

$C(5,4)3(-2)^4=5\times3\times16$。

M1	正确项
A1	系数

(b) · 3 marks

-1.

系数和为$F(2)$，即$(3-4)^5=-1$，不是$F(1)$。

M1	选择新多项式的$x=1$
M1	正确原输入2
A1	总和

Question 20

P2-C04-R20 · 5 marks

(a) · 3 marks

$a_2 = a_4 = 15$; $\binom{6}{2} = \binom{6}{4}$.

选择k个x与选择其余6−k个常数具有对称二项式系数。

M1	正确两组合数
A1	值
B1	对称性理由

(b) · 2 marks

14.

原三次20减原一次6，得14。

M1	正确幂次贡献
A1	系数

Question 21

P2-C04-R21 · 5 marks

(a) · 3 marks

$a = 2$.

一次12a，二次6a^2，等式除6a得a=2。

M1	两项系数
M1	使用非零条件
A1	a

(b) · 2 marks

16.

常数8，最高次系数a^3=8，相加16。

M1	正确两端系数
A1	总和

Question 22

P2-C04-R22 · 6 marks

(a) · 3 marks

$100^4 - 4(100)^3 + 6(100)^2 - 4(100) + 1$.

正整数四次展开，负1使符号交替。

M1	正确系数
M1	符号
A1	五项

(b) · 3 marks

601; each of the first three terms is a multiple of 1000, and $-400 + 1 = -399$ leaves final three digits 601.

减399等于减1000再加601，因此后三位601；明确处理负尾数。

M1	识别整千倍数
M1	处理负剩余
A1	三位数字

Question 23

P2-C04-R23 · 5 marks

(a) · 2 marks

$P(x) = (1 - x^2)^5$.

同次幂相乘可合为$[(1+x)(1-x)]^5$，再平方差。

M1	合并同次幂
A1	简式

(b) · 3 marks

10 and 0.

x^4来自选择两次$-x^2$，$C(5,2)=10$；只有偶次幂，x^7不存在。

M1	正确索引
A1	x四次系数
B1	奇次项为零

Question 24

P2-C04-R24 · 6 marks

(a) · 3 marks

$a = 2, b = 3$.

$a^4=16$取$a=2$；$4a^3b=96$给$32b=96$。

M1	常数确定正a
M1	一次项求b
A1	两值

(b) · 3 marks

$c = 216$; highest coefficient 81.

二次$6a^2b^2=6\times4\times9$，最高次$b^4=81$。

M1	二次项系数
A1	c
B1	最高次系数

Question 25

P2-C04-R25 · 6 marks

(a) · 3 marks

$1-14x+84x^2$.

系数1、7、21乘$(-2x)^k$。

M1	正确二项式系数
M1	负号和幂
A1	三项

(b) · 3 marks

0.8684.

设$1-2x=.98$得$x=.01$，代$1-.14+.0084=.8684$。

M1	正确选择x
M1	代展开
A1	估计

Question 26

P2-C04-R26 · 5 marks

(a) · 2 marks

270.

选两次x、三次3，$C(5,2)3^3$=270。

M1	正确项
A1	系数

(b) · 3 marks

$k=-3/2$.

来自原一次405与k倍原二次270，405+270k=0。

M1	识别两贡献
M1	正确一次系数
A1	k

Question 27

P2-C04-R27 · 6 marks

(a) · 4 marks

$n(n-1)/2-n=14$ gives $n=7$.

$n^2-3n-28=(n-7)(n+4)=0$，按正整数取7。

M1	正确差的顺序
M1	解二次方程
M1	正整数筛选
A1	n

(b) · 2 marks

35.

$C(7,3)=35$。

M1	正确组合数
A1	系数

Question 28

P2-C04-R28 · 6 marks

(a) · 3 marks

$48x^2 + 32x^4$.

常数2被减去，奇次项抵消，二次$2\times6\times4$，四次2×16。

M1	识别被抵消项
M1	两偶次系数
A1	多项式

(b) · 3 marks

$P(x) = 16x^2(3 + 2x^2) > 0$ for $x \neq 0$.

两个因子都严格正：x不为0使$16x^2$正，另因子至少3。

M1	因式分解
M1	两因子正性
A1	严格结论

Question 29

P2-C04-R29 · 6 marks

(a) · 2 marks

$a = 1$.

系数分别为$10a^2$和$5a$；$10a^2 = 2(5a)$，由$a > 0$取$a = 1$。

M1	正确两个二项系数
A1	合法参数

(b) For this value of a, the expansion of $(1 + ax)^5(1 + bx)$ has no x^3 term.

(i) · 2 marks

$b = -1$.

x^3系数为$10 + 10b$，令其为零得$b = -1$。

M1	两项贡献
A1	b

(ii) · 2 marks

-4.

x^5系数为$1 + 5b = 1 - 5 = -4$，不能忽略来自$bx \cdot 5x^4$的贡献。

M1	卷积的两项
A1	系数

Question 30

P2-C04-R30 · 7 marks

(a) · 3 marks

$128 + 480x^2 + 120x^4$.

展开中奇次项在两式中符号相反而抵消。偶次项：$2(2^6) = 128$，$2\binom{6}{2}2^4 = 480$，$2\binom{6}{4}2^2 = 120$。

B1	解释抵消
M1	二项系数
A1	三个非零项

(b) Let $Q(x) = (2+x)^6 - (2-x)^6$.

(i) · 2 marks

320.

差式保留并加倍奇次项，$2\binom{6}{3}2^3 = 320$。

M1	识别加倍奇次项
A1	系数

(ii) · 2 marks

440.

来自P的x^4项120及xQ的x^4项320，相加为440。

M1	追踪移位次数
A1	总系数